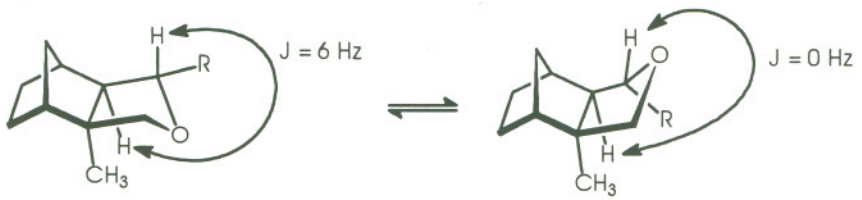
Organic Chemistry Cumulative Examination
April 23, 1994

1. Unlike six-membered rings, which strongly prefers a chair conformation, five-membered rings can adopt a variety of conformations with very small energy differences among them. Two types of five-membered ring conformations are called the "envelope" (four atoms in a plane, the fifth above or below) and the "half-chair" (three atoms in a plane, the fourth above, the fifth below)—you may think of them as being analogous to the chair and boat conformations of six-membered rings, but without their large energy differences.

When the five-membered ring is fused into a rigid skeleton, such as norbornane, it becomes locked into the envelope conformations, but should still be able to interconvert between the two. In the example shown below, the four carbon atoms of the five membered ring are co-planar, and the oxygen can be either above or below this plane. The two conformations are distinguishable from the magnitude of the NMR coupling constants between the two protons as shown.

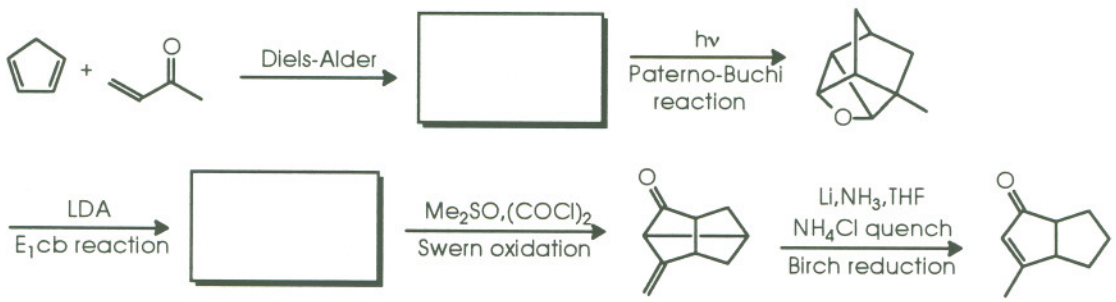


When R = H, it turns out that the two conformations are comparable in energy. Thus, there is no strong intrinsic preference of the five membered ring to adopt one conformation or the other. However, when R is an alkyl group (in the experiments, R = $CH_2CH_2C_6H_5$ and $CH_2COC_6H_5$), there is a definite preference for the conformation on the left. When R is an electronegative substituent (R = OH and $OCH_2C_6H_5$) there is a definite preference for the conformation on the right.

What is going on here? In particular, what term can you use to desribe the observed effect for the electronegative substitutents, and what is the explanation for this effect?

(Reference: *JACS* **1994**, *116*, 2340)

2. The following reaction sequence is, I think, a pretty smart way of obtaining the diquinane skeleton, which is found in many biologically active natural products. The relevant reaction types are indicated. You are to fill in the missing intermediates that belong in the empty boxes.



In addition, show in detail what happens in the last step (use electron pushing diagrams).

Some helpful information:

Paterno-Büchi reaction is a photochemical 2+2 cycloaddition between C=C and C=O.

LDA is lithium diisopropyl amide, a VERY strong base.

Swern oxidation converts alcohols into aldehydes or ketones under mild conditions. It uses dimethyl sulfoxide as the oxidant, along with oxalyl chloride, as noted above.

Birch reduction is generally introduced in textbooks in the context of reducing benzene to 1,4-cyclohexadiene. The mechanism involves transfer of an electron from lithium to the substrate, generating a radical anion, followed by protonation, transfer of another electron, and another protonation. Knowing this sequence of steps should help you figure out what happens in the reaction above.

I hope all of you already know what a Diels-Alder reaction is, as well as the designation E_1cb.

(Reference: *JACS* **1994**, *116*, 2613)